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# CONCEPT MAP: SYSTEM CAPABILITIES AND VISION-BASED APPLICATIONS
# Google Colab Code
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# Install Graphviz
!apt-get -qq install graphviz
!pip -q install graphviz

from graphviz import Digraph
from IPython.display import Image, display

# Create graph
dot = Digraph("SystemCapabilities", format="png")

# Layout settings
dot.attr(rankdir="TB")
dot.attr(size="16,16")
dot.attr(dpi="300")
dot.attr(nodesep="0.5")
dot.attr(ranksep="0.8")

# =====
# MAIN TOPIC
# =====

dot.node("A",
        "SYSTEM CAPABILITIES\nFOR\nVISION-BASED APPLICATIONS",
        shape="box",
        style="filled",
        fillcolor="lightblue",
        fontsize="22")

# =====
# SYSTEM CAPABILITIES
# =====

dot.node("B",
        "Processing Power\n(CPU / GPU)",
        shape="box",
        style="filled",
        fillcolor="lightgreen")

dot.node("C",
        "Memory Capacity",
        shape="box",
        style="filled",
        fillcolor="lightgreen")

dot.node("D",
        "Camera & Sensor Quality",
        shape="box",
        style="filled",
        fillcolor="lightgreen")

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dot.node("E",
        "Storage & Data Handling",
        shape="box",
        style="filled",
        fillcolor="lightgreen")

dot.node("F",
        "Network & Communication",
        shape="box",
        style="filled",
        fillcolor="lightgreen")

# =====
# EFFECTS
# =====

dot.node("G",
        "Real-Time Processing",
        shape="ellipse",
        style="filled",
        fillcolor="palegreen")

dot.node("H",
        "High Image Quality",
        shape="ellipse",
        style="filled",
        fillcolor="palegreen")

dot.node("I",
        "Fast Data Access",
        shape="ellipse",
        style="filled",
        fillcolor="palegreen")

dot.node("J",
        "Low Latency",
        shape="ellipse",
        style="filled",
        fillcolor="palegreen")

dot.node("K",
        "Reliable Performance",
        shape="ellipse",
        style="filled",
        fillcolor="palegreen")

# =====
# VISION PROCESSING
# =====

dot.node("L",
        "Image Acquisition",
        shape="box",
        style="filled",
        fillcolor="khaki")

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dot.node("M",
        "Image Preprocessing",
        shape="box")

dot.node("N",
        "Feature Extraction",
        shape="box")

dot.node("O",
        "Object Detection",
        shape="box")

dot.node("P",
        "Recognition & Decision Making",
        shape="box")

# =====
# APPLICATIONS
# =====

dot.node("Q",
        "Medical Imaging",
        shape="box",
        style="filled",
        fillcolor="lightskyblue")

dot.node("R",
        "Autonomous Vehicles",
        shape="box",
        style="filled",
        fillcolor="lightskyblue")

dot.node("S",
        "Face Recognition",
        shape="box",
        style="filled",
        fillcolor="lightskyblue")

dot.node("T",
        "Industrial Inspection",
        shape="box",
        style="filled",
        fillcolor="lightskyblue")

dot.node("U",
        "Surveillance Systems",
        shape="box",
        style="filled",
        fillcolor="lightskyblue")

# =====
# PERFORMANCE
# =====

dot.node("V",
        "Application Accuracy",
        shape="box",
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        style="filled",
        fillcolor="lightcyan")

dot.node("W",
        "System Efficiency",
        shape="box",
        style="filled",
        fillcolor="lightcyan")

dot.node("X",
        "User Reliability",
        shape="box",
        style="filled",
        fillcolor="lightcyan")

# =====
# LITERATURE
# =====

dot.node("Y",
        "Literature Perspectives\n\n"
        "• Gonzalez & Woods\nDigital Image Processing\n\n"
        "• Richard Szeliski\nComputer Vision\n\n"
        "• Forsyth & Ponce\nComputer Vision\n\n"
        "• Marr (Vision Theory)",
        shape="note",
        style="filled",
        fillcolor="lightyellow")

# =====
# CONNECTIONS
# =====

# Main
dot.edge("A", "B")
dot.edge("A", "C")
dot.edge("A", "D")
dot.edge("A", "E")
dot.edge("A", "F")

# Capabilities -> Effects
dot.edge("B", "G", label="enables")
dot.edge("C", "K", label="supports")
dot.edge("D", "H", label="improves")
dot.edge("E", "I", label="provides")
dot.edge("F", "J", label="reduces")

# Effects -> Processing
dot.edge("H", "L")
dot.edge("G", "M")
dot.edge("M", "N")
dot.edge("N", "O")
dot.edge("O", "P")

dot.edge("I", "P")
dot.edge("J", "P")
dot.edge("K", "P")

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# Processing -> Performance
dot.edge("P","V",label="improves")
dot.edge("P","W",label="increases")
dot.edge("P","X",label="enhances")

# Performance -> Applications
dot.edge("V","Q")
dot.edge("V","R")
dot.edge("V","S")
dot.edge("W","T")
dot.edge("X","U")

# Literature
dot.edge("Y","A",label="supported by")

# =====
# RENDER & DISPLAY
# =====

filename = dot.render("System_Capabilities_Concept_Map")

display(Image(filename))

print("Concept Map generated successfully!")
```

